

DeepCardio-RAG

Genuine Training Pipeline

Technology Stack, Datasets, Algorithms,
and Input / Process / Output Specification

Technical Specification Report

Companion to `DeepCardio_Genuine_Training.ipynb`

Provenance of the figures in this document. Architecture and hyperparameter values are read from the implementation source (`core/`, `colab_train_genuine.py`); performance values are read from `data/genuine_results.json` and `data/vfdb_cv_metrics.json`, which store held-out evaluations produced by the notebook itself. No value is copied from a manuscript draft. Where a paper draft and the code disagree, the *code* value appears in the body and the disagreement is recorded in Appendix A.

Reporting policy. Only genuinely trained modules on benchmark data are reported. Modules without trained weights are marked NOT TRAINED and carry no performance figures; no demonstration-patient or synthetic-signal output is presented as a result.

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1 Purpose and Scope

DeepCardio_Genuine_Training.ipynb is the single, current training notebook for the DeepCardio-RAG system. Its purpose is narrow and deliberate: train every module that can be trained on a *public benchmark dataset*, evaluate it on a *held-out* split, and persist both the weights and the measured metrics to disk. It exists to replace an earlier situation in which several modules ran on random-initialised weights while still emitting clinical-looking output.

This report documents four things, in the order requested:

1. the complete **technology stack** (Section 2);
2. the **datasets** used, with provenance, size, split protocol and known integrity issues (Section 3);
3. the **algorithms** implemented per module (Section 4);
4. the **input** → **process** → **expected output** specification for each module, including the observed held-out values (Section 5).

The notebook does *not* serve the web dashboard. The FastAPI application (`main.py`) is a separate local process; the notebook and the dashboard share only the artefacts written into `data/`.

2 Technology Stack

2.1 Runtime topology

The notebook is *dual-runtime*. Cells are tagged for one of two execution backends, and the two halves never communicate during a run.

Table 1: Execution backends and their responsibilities

Backend	Hardware	Responsibility
Colab cloud	NVIDIA T4 GPU, Google Drive mounted at <code>/content/drive</code>	All model training, dataset download, ablation studies. Cannot reach the private vector database.
Local runtime	Windows host that also runs the Milvus container; Jupyter exposed over <code>jupyter_http_over_ws</code>	Vector-database checks, retrieval grounding, pinned Milvus client.

The split works without any network tunnel because *no training module touches the vector database* — the training code contains no reference to `db_manager`, `init_db` or `pymilvus`. The hand-off between the two halves is Google Drive itself: weights written to `data/` on Colab synchronise back down to the local project folder. The largest artefact, `ecg_ptb_xl_encoder.pt`, is approximately 55 MB, so the local half must wait for that synchronisation to finish before it can confirm that the pipeline is using trained weights.

2.2 Software stack

Table 2: Software components and their role in the training pipeline

Layer	Component	Role
Deep learning	PyTorch (<code>torch</code> , <code>torch.nn</code> , <code>torch.optim</code>) <code>torch.utils.data</code>	All eight neural modules; CUDA on Colab, CPU fallback locally. <code>DataLoader</code> / <code>TensorDataset</code> / custom <code>Dataset</code> for the video and audio paths.

Table 2 continued

Layer	Component	Role
Language models	<code>transformers==4.44.2</code>	GPT-2 (124M) report decoder with soft-prompt injection; pinned because a newer build broke <code>GPT2LMHeadModel</code> import on Colab.
	<code>sentence-transformers</code>	Sentence-BERT <code>all-mpnet-base-v2</code> , 768-dim embeddings for retrieval.
Classical ML	<code>scikit-learn</code>	<code>RepeatedStratifiedKFold</code> , <code>StandardScaler</code> , <code>KNNImputer</code> , <code>GradientBoostingClassifier</code> , <code>RandomForestClassifier</code> , <code>ExtraTreesClassifier</code> , <code>LogisticRegression</code> , and all metrics (accuracy, macro- F_1 , ROC-AUC, Brier).
	<code>joblib</code>	Persists fitted scalers and feature-column lists alongside the network weights.
Signal / data I/O	<code>wfdb>=4.3.1</code>	Decodes PhysioNet WFDB waveforms and annotations (PTB-XL <code>.hea/.dat</code> , VFDB <code>.atr</code>). Version-critical — see Section 2.3.
	<code>pandas==2.2.3</code>	Tabular loaders (NHANES, UCI Cleveland, MIT-BIH CSV). Pinned by the <code>wfdb</code> requirement.
	<code>numpy (2.x)</code>	Array backbone throughout.
	<code>opencv-python-headless</code> <code>kagglehub</code>	Echo video decoding and frame resizing. Programmatic Kaggle dataset download (PTB-XL, NHANES).
Vector database	<code>pymilvus==2.6.9</code>	Client for Milvus standalone v2.6.2 at <code>localhost:19530</code> . Version-critical .
	Milvus standalone (Docker)	Collection <code>cardio_knowledge_base</code> : 768-dim, L_2 metric, IVF_FLAT index, <code>nlist</code> = 2048.
Serving (separate)	<code>fastapi</code> + <code>uvicorn</code>	REST API and static dashboard; not exercised by the notebook.
	<code>fpdf2</code>	Dual-audience PDF report generation.
Configuration	<code>pydantic-settings</code>	<code>config.py</code> settings object; real environment variables take precedence over <code>.env</code> .

2.3 Version pins that are load-bearing

Three pins in the dependency cell are not cosmetic. Each was added in response to a specific, diagnosed failure.

2.4 Notebook execution map

Two operational cautions carried in the notebook itself. (i) Cell 4 pins *Colab's* environment; it should not be re-run on the local runtime, where it would churn a working install. On a runtime switch, re-run cell 2 only. (ii) Section 5c must be skipped on a Colab cloud backend: a cloud VM cannot route to a private machine, and `db_manager` will silently fall back to FAISS while still logging **Vector DB ready**. That green message is not evidence of anything.

3 Datasets

Seven public corpora are used. Table 5 gives the summary; the subsections record provenance, split protocol and the integrity issues that were found and fixed.

Table 3: Load-bearing version pins

Pin	Failure it prevents
<code>wfdb>=4.3.1</code>	Under NumPy 2, older <code>wfdb.rdann()</code> raises <code>OverflowError: Python integer 256 out of bounds for uint8</code> . The VFDB loader then fell back to a hand-rolled byte parser whose SKIP handling byte-swapped the 32-bit interval, yielding annotation offsets up to 1.99×10^{10} on a 525,000-sample signal. Every window fell before the first valid annotation and was labelled Normal, producing a <i>single-class</i> dataset that scored accuracy 1.0 — a trivial solution, not a result.
<code>pandas==2.2.3</code>	Required by <code>wfdb 4.3.1</code> . Colab’s baseline is 2.2.2; a blanket <code>-upgrade</code> pulls pandas 3.0.x and breaks the pandas-heavy tabular pipelines. The resulting <code>google-colab 1.0.0</code> requires <code>pandas==2.2.2</code> warning is expected and harmless — it governs Colab’s own table display, not training.
<code>pymilvus==2.6.9</code>	<code>pymilvus 3.x</code> removes the 2.x ORM API (<code>connections / Collection / FieldSchema</code>) that <code>database/db_manager.py</code> is built on, and pulls <code>setuptools 83</code> , violating torch’s <code>setuptools<82</code> requirement.
<code>torch</code> (not up-graded)	Colab’s preinstalled CUDA build works; upgrading it breaks <code>transformers</code> model imports (e.g. <code>Could not import GPT2LMHeadModel</code>).

Table 4: Cell-to-runtime map (0-indexed over all cells, markdown included)

Cell	Runtime	Function
2	either	Runtime detection; mounts Drive on Colab, sets <code>MILVUS_HOST/MILVUS_PORT</code> locally; inserts CWD on <code>sys.path</code> .
4	Colab	Dependency install with the pins of Table 3.
5	either	In-kernel version assertion (pip output alone is not proof the kernel took the upgrade).
7	Colab	Kaggle credentials.
9	Colab	PTB-XL completeness fix; forces a full download to local disk and repoints the loader.
11	Colab	PhysioNet fetch of VFDB and CirCor via <code>wget -r -N -c</code> ; completeness check against <code>RECORDS</code> .
13	Colab	Optional SHA256 byte-level integrity check.
15	Colab	<code>run_all()</code> — trains every module in sequence.
17–19, 21	Colab	Individual modules: PTB-XL flagship, arthritis, MIT-BIH arrhythmia, CardioFusion.
20	local	Pins <code>pymilvus</code> to 2.6.9.
23–26	Colab	Echo, VFDB, PCG, heart disease.
28–29	local	Milvus connection and schema check; confirms the project is on the real server and not a silent fallback.
31	local	Confirms the pipeline loads the trained encoder and that retrieval is diagnosis-grounded.
33	either	RAG retrieval evaluation (in-memory).
35	Colab	Ablation studies.
37	Colab	VFDB recording-level cross-validation.

3.1 PTB-XL — flagship 12-lead ECG

Clinical 12-lead ECGs at 500 Hz, 10 seconds per record (12×5000 samples). The loader targets the five diagnostic superclasses {NORM, MI, STTC, CD, HYP} and uses the official patient-independent folds, so no patient appears in more than one split.

PTB-XL assigns *multi-label* superclasses, so the published per-class record counts overlap rather than partition the corpus: NORM 9 528, MI 5 486, STTC 5 250, CD 4 907, HYP 2 655 over 21 837 records from 18 885 patients. These are reference frequencies stored in the loader as `PTBXL_REFERENCE_COUNTS`; the training run samples a subset (3 000 records by default) and reports metrics on its own held-out fold.

Table 5: Datasets used by the genuine training pipeline

Module	Dataset		Size used		Classes	Split protocol
Flagship ECG	PTB-XL 500 Hz (PhysioNet)		3 000 sampled (2 389/296/315)	records	5 super-classes	Official folds 1–8/9/10, patient-independent
ECG arrhythmia	MIT-BIH beats (Kaggle CSV)	AAMI	2 999 train / 21 892 test		5 (N,S,V,F,Q)	Official train/test CSV split
CardioFusion	MIT-BIH, ECG-signal path only		2 549 train / 21 892 test		5	Train/val carve-out; test scored once
Ventricular	MIT-BIH (PhysioNet)	VFDB	22 recordings, 5 s windows		2 binary + 4 rhythm	Whole recordings per split; 2×5 -fold recording-level CV
Heart sound	CirCor (PhysioNet)	DigiScope	942 subjects, 2 057/480/626 recordings		3 murmur	Subject-level (patient-independent)
Heart disease	UCI Cleveland, Kaggle mirror		1 025 rows → 302 unique		2	5×3 repeated stratified K-fold + 25% hold-out
Arthritis	NHANES (CDC/Kaggle)	MCQ	5 756 records (3 000 sampled)		2	5×3 repeated stratified K-fold + 20% hold-out
Echo (LVEF)	EchoNet-Dynamic (Stanford AIMI)		not obtained		3 EF categories	n/a — access required

Integrity note. Drive-cached copies of PTB-XL are sometimes missing waveform folders, and a missing record silently falls back to a *synthetic* signal — which would contaminate results with fabricated data. Cell 9 exists solely to prevent this: it force-downloads a complete copy to local Colab disk and reprints the loader. The acceptance check is that `records500/21000` contains approximately 838 `.hea` files; a count of 0 means the mirror itself is incomplete.

3.2 MIT-BIH Arrhythmia — beat classification

Pre-segmented single-beat vectors of 187 samples, distributed as CSV, with the five AAMI classes N, S, V, F, Q. The official train/test CSV split is used; the test set of 21 892 beats is many times larger than the 2 999 sampled training beats, which makes the reported accuracy pessimistic relative to a balanced protocol. The class distribution is strongly imbalanced toward N, so macro- F_1 and AUC are the honest headline figures and raw accuracy is not.

3.3 MIT-BIH VFDB — ventricular arrhythmia

Twenty-two dual-channel recordings, 30 minutes each at 250 Hz, segmented into 5-second windows. Rhythms {VT, VF, VFL, VFIB, ASYS, HGEA, VER} are treated as *dangerous*; the remainder (N, AFIB, NOISE, SBR, SVTA, NOD, B, PM) form the background class. Splits are by whole recording, so a recording never contributes windows to two splits.

Integrity note. Twenty-two recordings cannot support a point estimate from a single 15/3/4 split. The single split gave test F_1 (dangerous) 0.5147 / AUC 0.8046 against validation 0.894 / 0.984 — a gap over four test recordings. It was therefore replaced by 2×5 -fold recording-level cross-validation (cell 37), and the cross-validated figures are the reportable ones.

3.4 CirCor DigiScope — phonocardiogram murmur

Heart-sound recordings at 4 000 Hz from multiple auscultation locations (AV, PV, TV, MV, Phc), labelled {Absent, Present, Unknown}. *Unknown* is a genuine annotation meaning auscultation was inconclusive —

not a missing-data marker. The public training subset is 3 163 recordings from 942 subjects; the full release (5 272 recordings, 1 568 subjects) retains a hidden test set.

Integrity notes, both fixed. (i) Subject identifiers were parsed with `rsplit("_", 1)`, which splits repeat-recording names such as `50204_MV_1.wav` on the *last* underscore, yielding patient id `50204_MV` instead of `50204`. This created 20 phantom subjects (962 counted against 942 real), silently defaulted their murmur label to the *Unknown class*, and — most seriously — allowed one patient to appear in both train and test. Corrected to `partition("_")`. (ii) A recording truncated by an interrupted download (`85286_AV.wav`, 15 974 of 161 280 bytes) was repaired; the download command uses `wget -N -c` and never `-nc`, which skips any file that merely exists without checking completeness.

3.5 UCI Cleveland — heart disease

The Kaggle mirror ships 1 025 rows, but these are the 303-record Cleveland cohort expanded with duplicate rows. Identical rows on both sides of a random split inflate every metric, so training deduplicates to the **302 unique** records first.

Integrity note. This mirror’s `target` column is inverted relative to the module’s own documentation: measured on the file, the `target==0` group carries the diseased clinical profile. The prediction path detects the polarity from clinical priors rather than hardcoding a flip, so a corrected upstream file would be left alone.

3.6 NHANES — arthritis risk

CDC National Health and Nutrition Examination Survey, Medical Conditions Questionnaire: 5 756 records (2 752 male, 3 004 female, mean age 49.1 years). The prediction target is **MCQ160A**, “has a doctor ever told you that you have arthritis” — self-reported, doctor-diagnosed, predominantly osteoarthritis. It is *not* serological rheumatoid arthritis, and results are not comparable with RA-cohort literature.

Integrity note. The arthritis-type field MCQ195 (RA/OA) is a skip-pattern question asked only of arthritis-positive respondents; it is excluded as a target-leakage source. Retaining it inflated AUC from an honest ≈ 0.79 to ≈ 0.99 — the ablation study that revealed this is cell 35.

3.7 EchoNet-Dynamic — not obtained

Apical 4-chamber echocardiogram videos with EF, ESV and EDV annotations. Access requires registration with Stanford AIMI and the dataset is **not present**; `genuine_results.json` records an **error** entry for this module rather than a metric. The echo module therefore has **no trained weights**, and every figure that would describe echo performance is withheld throughout this report.

4 Algorithms

All architectural values below are read from the implementation.

4.1 Flagship encoder — ECGTransformerMoE

Defined in `core/pipeline.py`; trained by `core/train_ptb1.py` through the wrapper `ECGSuperclassNet`.

1. **Patch embedding:** `Conv1d(12, 256, kernel=50, stride=50)` maps the 12×5000 input to 100 non-overlapping patches of 256 dimensions — one patch per 100 ms at 500 Hz.
2. A learnable `[CLS]` token is prepended, giving $(B, 101, 256)$, and learnable positional embeddings are added.
3. **6 Transformer blocks**, each with 8-head self-attention and a **mixture-of-experts feed-forward** layer: 4 experts, top-2 routing, FFN dimension 1024, dropout 0.1.
4. The `[CLS]` output passes through LayerNorm and a $256 \rightarrow 256$ projection to yield the ECG embedding.
5. A diagnostic head maps the embedding to the 5 PTB-XL superclasses. The embedding is separately projected $256 \rightarrow 768$ to query the vector store.

Total parameters: 14 453 765. Optimiser Adam, learning rate 3×10^{-4} , batch size 32, 30 epochs, seed 42. A ResNet-34 1D encoder is available as the ablation baseline (`ablate_ecg_encoder`).

4.2 Beat classifier — ECGArrhythmiaCNN

Four 1D-convolution blocks with channel widths (32, 64, 128, 256) and kernels (7, 5, 3, 3), each followed by batch normalisation, ReLU and pooling; then a fully connected head $256 \rightarrow 128 \rightarrow 64 \rightarrow 5$. The input tensor is single-channel, $(N, 1, 187)$. Class imbalance is handled with a weighted cross-entropy loss.

4.3 Ventricular detector — VentricularArrhythmiaDetector

A dual-channel 1D-CNN (`in_channels=2`) over 5-second windows, with *two heads*: a binary dangerous-vs-background head and a 4-class rhythm-family head {Normal, VT, VF/VFL, Other_Dangerous}. 313 958 parameters. The class carries an explicit `is_trained` flag, set true only when genuine weights are loaded, so that a random-weight detector cannot surface a live alert.

Cross-validation (`core/cv_vfdb.py`) folds *whole recordings*, stratified by dangerous-window fraction; validation recordings are drawn from the training pool only, and each test fold is scored exactly once after early stopping. Single-class test folds report AUC as NaN and are counted rather than silently averaged; a single-class *training* fold raises.

4.4 Murmur classifier — HeartSoundClassifier

Audio at 4000 Hz is cut into 5-second segments and converted to log-Mel spectrograms with 64 Mel bands and hop length 128, padded or cropped to a fixed width of 160 frames for batching. A 2D-CNN with three convolution blocks (16, 32, 64) and adaptive average pooling feeds a 3-class softmax head. 44 707 parameters — by far the smallest network in the system.

4.5 Tabular BERT + Mixture-of-Experts

Two instances of the same design, with *different* hyperparameters:

Table 6: TabularBERT-MoE instances as implemented

Instance	d_{model}	Heads	Layers	Experts
Arthritis (TabularBERTMoE)	64	4	3	6
Heart disease (HeartDiseaseBERTMoE)	64	4	2	4

Each scalar feature value is projected to d_{model} and summed with a learned feature-identity embedding, so the feature set is treated as a token sequence; a Transformer encoder then contextualises the tokens and a gated mixture-of-experts head produces the class logits.

For arthritis the network is one input to a **stacking ensemble**: Gradient Boosting, Random Forest and Extra Trees form the base layer alongside the BERT-MoE, and a logistic-regression meta-learner is fitted on out-of-fold predictions. Minority-class balancing (SMOTE-style oversampling), imputation and scaling are all fitted *inside* each training fold, never across the split.

For heart disease, plain logistic-regression and gradient-boosting baselines are evaluated on *identical folds*, so the added complexity of the deep model can be judged against them.

4.6 CardioFusion

A cross-modal Transformer with a multi-gate mixture-of-experts head over six tasks. Multi-task loss weights are *learned* by homoscedastic uncertainty weighting (Kendall et al.), not hand-set. Because no public benchmark provides same-patient four-modality paired data, only the **ECG-signal path** is trained, on MIT-BIH beats resampled to a 512-sample segment length. The remaining heads and the joint contrastive alignment are untrained — visible in the learned task weights, where only the arrhythmia task has moved off its 0.5 initialisation (to 0.4196).

4.7 Retrieval subsystem

Retrieval is *diagnosis-grounded*: the query is built from the predicted PTB-XL superclass, not from a raw embedding, and if the diagnostic head is absent the pipeline deliberately serves labelled placeholders rather than querying with something unfounded.

Three rankers are evaluated over a 67-document clinical corpus with 38 labelled queries: a lexical TF-IDF baseline, a semantic Sentence-BERT ranker (**all-mpnet-base-v2**, 768-dim), and a **hybrid reciprocal-rank-fusion** ranker combining the two. Metrics are Recall@ k , Precision@ k , MRR, nDCG@ k and Hit@ k , reported overall and per superclass.

5 Input, Process and Expected Output

This section is the operational specification: for each module, what goes in, what the algorithm does, and what should come out — both in *form* (the shape and keys of the result) and in *value* (the held-out figures actually observed).

How to read the “expected value” column. These are the measurements stored in `data/genuine_results.json` from the runs that produced the current weights. A re-run on the same seed and record count should land close to them; a substantially *higher* figure is a reason to check for leakage, not a success.

5.1 Module 1 — Flagship ECG superclass encoder

Entry point	<code>train_ptbxl(n_records=3000, epochs=30)</code> — notebook cell 17
Input	PTB-XL WFDB records; tensor ($B, 12, 5000$) — 12 leads, 10 s at 500 Hz. Labels: 5 diagnostic superclasses. Patient-independent official folds 1–8 (train), 9 (val), 10 (test).
Process	Patch embedding (100 patches of 50 samples) $\rightarrow$ [CLS] + positional embeddings $\rightarrow$ 6 Transformer blocks with 4-expert top-2 MoE FFN $\rightarrow$ LayerNorm $\rightarrow$ 256-dim embedding $\rightarrow$ diagnostic head. Adam, lr 3×10^{-4} , batch 32, 30 epochs; model selection on validation macro- F_1 ; test scored once.
Output artefacts	<code>data/ecg_ptbxl_encoder.pt</code> (≈ 55 MB, loaded by <code>core/pipeline.py</code>), <code>data/ecg_ptbxl_metrics.json</code>
Expected values	accuracy 0.5778 · macro- F_1 0.4866 · macro-AUC (OvR) 0.8007 · Brier 0.5966 · ECE 0.117. Per-class AUC: NORM 0.8832, MI 0.7817, STTC 0.8380, CD 0.7432, HYP 0.7572. Best validation macro- F_1 0.5418. 14.45M parameters, 209.4 s on CUDA.

5.2 Module 2 — MIT-BIH beat arrhythmia classifier

Entry point	<code>train_ecg_arrhythmia(n_records=3000, epochs=25)</code> — notebook cell 19
Input	MIT-BIH beat CSV; tensor ($N, 1, 187$). Labels: 5 AAMI classes (N, S, V, F, Q). Official train/test split.
Process	Four 1D-conv blocks (32, 64, 128, 256) with batch norm and pooling $\rightarrow$ FC 256 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 5. Class-weighted cross-entropy for the heavy N-class imbalance.
Output artefacts	<code>data/ecg_arrhythmia_cnn_model.pt</code> , <code>data/ecg_arrhythmia_scaler.pkl</code>
Expected values	accuracy 0.6065 · macro- F_1 0.5086 · macro-AUC (OvR) 0.9283 , on 21 892 held-out beats. Quote AUC and macro- F_1 ; accuracy is depressed by the 2 999/21 892 train-to-test ratio.

5.3 Module 3 — Ventricular arrhythmia detector (VFDB)

Entry point	<code>train_vfdb(epochs=30)</code> — cell 24; <code>cross_validate_vfdb(n_folds=5, n_repeats=2, epochs=30)</code> — cell 37
Input	22 VFDB recordings, dual-channel, 250 Hz, cut into 5-second windows ($B, 2, 1250$). Labels: binary dangerous-vs-background plus 4-class rhythm family.
Process	1D-CNN encoder with two heads. Folds are whole recordings stratified by dangerous-window fraction; validation drawn from the training pool; each test fold scored once after early stopping on validation F_1 (dangerous). Training raises if any split is single-class.
Output artefacts	<code>data/vfdb_detector.pt</code> , <code>data/vfdb_metrics.json</code> , <code>data/vfdb_cv_metrics.json</code>
Expected values	Cross-validated (reportable): AUC 0.9187 [0.8466, 0.9907] · F_1 (dangerous) 0.7184 [0.6217, 0.8151] · accuracy 0.8672 [0.8127, 0.9216] · rhythm macro- F_1 0.4512 [0.3831, 0.5193], over 10 fits, 176.8 s, 0 single-class test folds. Observed fold range AUC [0.6414, 0.9833]. Single split (provenance only): AUC 0.8046, F_1 (dang.) 0.5147.

Interval caveat, to be quoted with the number. The 95% figure is a t -interval over fold scores. Folds share training recordings, so the scores are not independent and the interval is *optimistic*: it describes spread across folds, not a confidence interval over patients. The superseded single split scored 0.8046, which lies *below* the interval low of 0.8466 — direct evidence that the interval is too narrow. The rhythm head (0.4512) must not be quoted as a strength; the binary head is the safety-relevant output.

5.4 Module 4 — Heart sound murmur (CirCor)

Entry point	<code>train_circor(n_max=3000, epochs=25)</code> — notebook cell 25
Input	CirCor .wav at 4000 Hz → 5-second segments → log-Mel spectrogram ($B, 1, 64, 160$). Labels: {Absent, Present, Unknown}. Subject-level split, 2057/480/626 recordings over 942 subjects.
Process	2D-CNN (16, 32, 64) with batch norm and adaptive average pooling → 3-class softmax. Early stopping on validation macro- F_1 ; test scored once.
Output artefacts	<code>data/heart_sound_classifier.pt</code> , <code>data/circor_metrics.json</code>
Expected values	accuracy 0.7604 · macro- F_1 0.5347 · Present-vs-Absent AUC 0.7611 · per-class F_1 : Absent 0.8616, Present 0.4776, Unknown 0.2651. 44 707 parameters, 1 249.3 s.

Acceptance checks for a re-run (all three passed on the current weights): the loader must report **942 subjects**, not 962 — 962 means a stale loader is cached and the kernel needs restarting; *no* recordings skipped for a missing metadata row; and Present-vs-Absent AUC *below* 0.7981, because 0.7981 is the leakage-contaminated figure and leakage inflates. Validation macro- F_1 is unstable across epochs, so quote the AUC as the headline and do not present macro- F_1 as precise.

5.5 Module 5 — Heart disease risk (UCI Cleveland)

The baseline comparison is the point of this module: on 302 unique records a plain logistic regression beats the TabularBERT-MoE on every metric including calibration. That result should be reported as it stands.

5.6 Module 6 — Arthritis risk (NHANES)

Two caveats to carry with these numbers. First, the three leading features are collinear encodings of age, together roughly 40% of total importance — this is substantially an age model. Second, the hold-out AUC (0.8127) lies *outside* the cross-validated interval [0.7813, 0.8038]; the cross-validated figure is the headline. A run returning AUC near 0.99 indicates MCQ195 has re-entered the feature set.

Entry point	<code>train_heartdisease(n_repeats=5, n_splits=3, epochs=80)</code> — notebook cell 26
Input	<code>data/heart.csv</code> deduplicated 1025 → 302 unique rows; 13 clinical features expanded to 21 after encoding. Binary target.
Process	5 × 3 repeated stratified K-fold with KNN imputation and scaling fitted inside each fold. TabularBERT-MoE (2 layers, 4 experts) trained per fold, plus logistic-regression and gradient-boosting baselines on identical folds. 25% hold-out evaluated once.
Output artefacts	<code>data/heart_disease_bert_moe_model.pt</code> , <code>data/heart_disease_bert_moe_scaler.pkl</code> , <code>data/heart_disease_metrics.json</code>
Expected values	BERT-MoE: accuracy 0.7967 [0.7797, 0.8136], AUC 0.8718 [0.8594, 0.8841], F_1 0.7939, Brier 0.1749. Logistic baseline: AUC 0.8921 [0.8861, 0.8980] — higher than the deep model. GBT: AUC 0.8833. Hold-out (25%) BERT-MoE: accuracy 0.7763, AUC 0.8551.

Entry point	<code>train_arthritis(n_records=3000)</code> — notebook cell 18
Input	NHANES MCQ; 10 base predictors (Age, Gender_M, PovertyRatio, EducationLevel, BMI, SystolicBP, DiastolicBP, HasGout, HasOsteoporosis, InflammationProxy) expanded to 17 with engineered terms (<code>age_risk</code> , <code>age_squared</code> , <code>bmi_squared</code> , <code>obesity_flag</code> , <code>age_gender</code> , <code>pulse_pressure</code> , <code>map</code>). Target MCQ160A.
Process	5 × 3 repeated stratified K-fold with imputation, scaling and SMOTE-style balancing fitted <i>inside</i> folds. Base layer GBM + RF + ExtraTrees + TabularBERT-MoE (3 layers, 6 experts); logistic-regression meta-learner on out-of-fold predictions. 20% stratified hold-out taken before balancing.
Output artefacts	<code>data/arthritis_bert_moe_model.pt</code> , <code>data/arthritis_bert_moe_scaler.pkl</code> (carries the <code>feature_cols</code> list)
Expected values	CV accuracy 0.7622 [0.7540, 0.7704] · CV AUC 0.7925 [0.7813, 0.8038] · CV macro- F_1 0.6844 · CV Brier 0.1624. Hold-out: accuracy 0.7700, AUC 0.8127, F_1 0.7024. Top features: <code>age_risk</code> 0.157, <code>age_squared</code> 0.128, Age 0.115.

5.7 Module 7 — CardioFusion (ECG path)

Entry point	<code>train_cardiofusion(n_records=3000, epochs=20)</code> — notebook cell 21
Input	MIT-BIH beats reshaped to 512-sample ECG segments; 2549 train / 21892 test. Only the ECG-signal modality is populated.
Process	Cross-modal Transformer with multi-gate MoE head over six tasks; loss weights learned by homoscedastic uncertainty weighting. Model selection on a validation split carved from train; class-weighted arrhythmia loss; test scored once.
Output artefacts	<code>data/cardiofusion_weights.pt</code> , <code>data/cardiofusion_metrics.json</code>
Expected values	accuracy 0.8944 · macro- F_1 0.5401 · AUC 0.8384. Learned task weights: arrhythmia 0.4196 , all five other tasks still at their 0.5 initialisation.

Those five unchanged weights are the diagnostic fact: the corresponding heads never received gradient. Only the arrhythmia output of this module is a trained result; the EF, murmur, ventricular-alert and rhythm heads must not be surfaced, and no composite score aggregating them is meaningful. The run is also unstable across epochs, so a single run is not a defensible point estimate.

5.8 Module 8 — LVEF estimation (blocked)

Entry point	<code>train_echonet(n_train=3000, n_eval=1000, epochs=25)</code> — notebook cell 23
Input	EchoNet-Dynamic A4C videos, 32 frames at 112×112 grayscale. Dataset not present.
Process	3D-CNN video encoder (<code>EchoNetModel</code> , 384-dim embedding) $\rightarrow$ EF regression head, with predictions clamped to $[0, 100]\%$, plus a 3-way EF category head (HFrEF < 40 , HFmrEF 40–50, Normal ≥ 50).
Expected output	None. The module records an error entry in <code>genuine_results.json</code> and produces no weights. Consequently the serving layer returns <code>ef_predicted: null</code> with <code>model_trained: false</code> rather than a number.

5.9 Retrieval evaluation

Entry point	<code>run_rag_evaluation(include_semantic=True)</code> — notebook cell 33
Input	67-document clinical guideline corpus; 38 labelled queries with per-superclass relevance annotations.
Process	Three rankers scored on identical queries: lexical TF-IDF, semantic Sentence-BERT (<code>all-mpnet-base-v2</code>), and hybrid reciprocal rank fusion. Metrics $\text{Recall}@k$, $\text{Precision}@k$, MRR, $\text{nDCG}@k$, $\text{Hit}@k$, reported overall and per superclass.
Expected values	$\text{Recall}@5$: hybrid RRF 0.939 versus lexical TF-IDF 0.899. Retrieval runs top- k with $k = 5$ against a 67-document corpus at 768 dimensions under L_2 .

Because the corpus is small and hand-seeded, these figures verify that the retrieval *mechanism* works; they are not evidence of retrieval quality at realistic corpus scale.

5.10 Grounding check (local runtime)

Cell 31 is the acceptance test for the whole chain. It reloads `core/pipeline.py`, calls `get_model()` and asserts three things: the encoder reports `encoder_is_trained`, the PTB-XL diagnostic head is loaded, and the retrieval source is a member of `GROUNDED_RETRIEVAL_SOURCES` (`hybrid-rrf` or `milvus-sbert`, *not* `demo-fallback`). Membership is tested rather than equality against a literal, so that a change of retrieval strategy does not silently report correctly-retrieved documents as ungrounded.

6 Consolidated Results

What this table does not say. No module here exceeds its published benchmark. The flagship encoder at AUC 0.8007 sits below the ≈ 0.92 reported for PTB-XL in the literature; the murmur classifier at macro- F_1 0.5347 is below the CirCor challenge figures; the heart disease deep model is beaten by logistic regression on its own folds. Report-generation quality (BLEU, factuality) has never been measured and no figure for it should be quoted.

7 Reproduction Procedure

1. **Colab, T4 GPU.** Run cell 2 (mounts Drive, detects runtime), then cell 4 (dependencies), then cell 5 and confirm in-kernel `pandas 2.2.3 | wfdb 4.3.1 | numpy 2.0.2`. Pip’s own output is not sufficient evidence — a pip upgrade lands underneath a kernel that has already imported the old version.
2. Cell 7 (Kaggle credentials), cell 9 (PTB-XL completeness), cell 11 (VFDB + CirCor). Datasets already on Drive are not re-downloaded; `wget -N -c` skips what is current.

Table 7: Held-out results, all modules (source: `data/genuine_results.json`)

Module	Dataset	Accuracy	Macro- F_1	AUC	Trained
Flagship ECG	PTB-XL	0.5778	0.4866	0.8007	yes
Beat arrhythmia	MIT-BIH	0.6065	0.5086	0.9283	yes
Ventricular (CV)	VFDB	0.8672	0.7184 [†]	0.9187	yes
Heart sound	CirCor	0.7604	0.5347	0.7611 [‡]	yes
Heart disease	UCI Cleveland	0.7967	0.7939	0.8718	yes
Arthritis	NHANES	0.7622	0.6844	0.7925	yes
CardioFusion	MIT-BIH (ECG)	0.8944	0.5401	0.8384	partial
LVEF	EchoNet	—	—	—	no
RAG retrieval	67 docs / 38 queries	Recall@5 0.939 (hybrid RRF) vs 0.899 (lexical)			yes

[†] F_1 (dangerous), the safety-relevant binary head. [‡] Present-vs-Absent AUC.

CardioFusion: only the arrhythmia head is trained; five task heads remain at initialisation.

- Either cell 15 (`run_all`) or the individual module cells 17–19, 21, 23–26. Every step is guarded: a missing dataset records an **error** entry and the remaining modules continue.
- Cell 37 for VFDB cross-validation (10 fits, ≈ 177 s).
- Cell 35 for the ablation studies.
- Wait for Drive to finish syncing `data/` back down.
- Switch to the local runtime**, re-run cell 2 *only*, then cell 20 (pinned client), cells 28–29 (Milvus checks), cell 31 (grounding check), cell 33 (RAG evaluation).

Prerequisites on the local host: Docker Desktop running with the `milvus-standalone` container started (it does not auto-start after a reboot), and the Jupyter bridge installed (`jupyter_http_over_ws` plus `notebook`).

A Paper-versus-Code Specification Deltas

While assembling this document from source, the following architectural values in `DeepCardio_CompIntel_Submit.tex` were found to differ from the implementation. They are recorded here rather than silently reconciled; the *code* column is what actually ran to produce the results in Table 7.

Item	Paper states	Code implements	Source
Heart-disease BERT-MoE	4-layer BERT, 6-expert MoE	2 layers, 4 experts	<code>heart_disease_pipeline.py:289</code>
Arthritis BERT-MoE	4-layer BERT, 6 experts	3 layers , 6 experts	<code>arthritis_pipeline.py:704</code>
PCG Mel bands	128 Mel bins	64	<code>heart_sound_loader.py:47</code>
PCG hop length	hop 256	128	<code>heart_sound_loader.py:48</code>
PCG window	3-second, 50% overlap	5-second segments	<code>heart_sound_loader.py:46</code>
Echo video depth	$64 \times 112 \times 112$	$32 \times 112 \times 112$	<code>echonet_loader.py:27</code>
Beat-classifier input	“two leads: Lead II + modified V1”	single-channel tensor ($N, 1, 187$)	<code>ecg_arrhythmia_pipeline.py:185</code>

A separate stale descriptor exists in the code itself: `arthritis_pipeline.py:1127` still builds the string “Hybrid Stack: BERT-MoE + GBM + RF + ExtraTrees + SVC $\rightarrow$ LogReg Meta-Learner”, although the fitted base learners are only `{gbm, rf, et}`. The same phantom SVC was removed from the paper and from the dashboard; this occurrence was not part of either fix.

B Artefacts Written to `data/`

File	Written by / consumed by
<code>ecg_ptbxxl_encoder.pt</code>	<code>train_ptbxxl</code> → <code>core/pipeline.py</code> (flagship inference)
<code>ecg_ptbxxl_metrics.json</code>	<code>train_ptbxxl</code>
<code>ecg_arrhythmia_cnn_model.pt</code> , <code>ecg_arrhythmia_scaler.pkl</code>	<code>train_ecg_arrhythmia</code>
<code>vfdb_detector.pt</code> , <code>vfdb_metrics.json</code>	<code>train_vfdb</code>
<code>vfdb_cv_metrics.json</code>	<code>cross_validate_vfdb</code>
<code>heart_sound_classifier.pt</code> , <code>circor_metrics.json</code>	<code>train_circor</code>
<code>heart_disease_bert_moe_model.pt</code> , <code>heart_disease_bert_moe_scaler.pkl</code> , <code>heart_disease_metrics.json</code>	<code>train_heartdisease</code>
<code>arthritis_bert_moe_model.pt</code> , <code>arthritis_bert_moe_scaler.pkl</code>	<code>train_arthritis</code> ; the scaler carries <code>feature_cols</code>
<code>cardiofusion_weights.pt</code> , <code>cardiofusion_metrics.json</code>	<code>train_cardiofusion</code>
<code>genuine_results.json</code>	<code>run_all</code> — the single source of truth for every reported metric

Standing rule. `data/genuine_results.json` is the only sanctioned source of performance figures for this system. A number that cannot be traced to it, or to the raw metrics files it summarises, should not appear in a paper, a dashboard tile, or a report.